Deep Learning Fundamentals

Synthetic educational document created for RAG development and evaluation practice. It is intended as a controlled development dataset, not as an authoritative source.

1. What is Deep Learning?

Deep learning is a family of machine learning methods based on neural networks with multiple computational layers. These networks can learn increasingly complex representations from data. Deep learning has been applied to image recognition, speech processing, natural language processing, recommendation systems, and many other tasks.

2. Artificial Neurons

An artificial neuron receives numerical inputs, multiplies them by learnable weights, adds a bias, and applies an activation function. A simplified neuron can be written as y = activation(w1x1 + w2x2 + ... + b). Multiple neurons are arranged into layers to form a neural network.

3. Activation Functions

Activation functions introduce nonlinearity into neural networks. ReLU, or rectified linear unit, outputs zero for negative inputs and the input itself for positive inputs. Sigmoid maps a value into a range between zero and one and is often used in binary classification output layers. Softmax converts a vector of scores into a probability distribution over multiple classes.

4. Backpropagation

Backpropagation is an algorithm for computing gradients of a loss function with respect to neural network parameters. During training, a forward pass produces predictions and a loss value. Backpropagation applies the chain rule to calculate gradients, and an optimizer such as stochastic gradient descent updates the weights.

5. Loss Functions and Optimization

A loss function measures how different model predictions are from the target values. Mean squared error is common for regression, while cross-entropy is common for classification. Optimization algorithms use gradients to reduce the loss. Learning rate controls the size of parameter updates and is one of the most important training hyperparameters.

6. Regularization

Neural networks can overfit, especially when the network is large relative to the amount of training data. Dropout randomly disables a fraction of neurons during training. Weight decay penalizes large weights. Early stopping stops training when validation performance stops improving. Data augmentation creates modified training examples and is particularly useful for image tasks.

7. Convolutional and Recurrent Networks

Convolutional neural networks use convolutional filters to capture local patterns and have historically been important for image processing. Recurrent neural networks process sequences while maintaining a hidden state. LSTM and GRU architectures were designed to improve the ability of recurrent networks to retain information over longer sequences.